MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
4000 PROCESS PLANT: WEIGHBRIDGE CONTROL ROOM
DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Weighbridge Control Room.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Weighbridge Control Room Building only.

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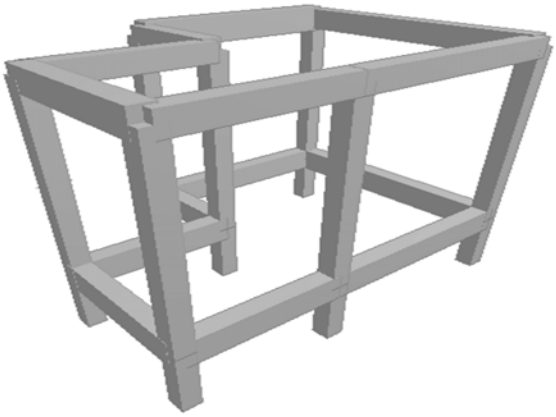
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4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-CAL-1008-A02.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

5. DESIGN LOADS

5.1. Dead Load(DL)

5.1.1. Roof Dead Loads

• Provision for Electrical Lighting	= 0.05 kN/m²
Ceiling	
Suspended steel channel system	= 0.10 kN/m²
Acoustical Fiber Tile (16mm)	= 0.05 kN/m²
Sub-total	= **0.2 kN/m²**

- Hollow Core Slab

  (HCS-150) Self Weight= **2.52 KN/m2**

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Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m^2 for Finishes Characteristic Service Loads Kn/m^2							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/ m²**

Total DL =3.1+0.2+2.52 = **5.82 KN/m2**

TOTAL DL as UDL ON BEAMS

- UDL on LB's Along Grid Line-A & C between 2 & 3 = 5.82 x 5.10/2 = **14.85 Kn/m**
- UDL on LB's Along Grid Line-B & C between 1 & 2 = 5.82 x 3.0/2 = **8.70 Kn/m**

Moment on all the External Beams due to Cantilever Portion of Steel Canopy = **2 knm/m**

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5.1.2. Ground Floor Walls

- 250 tyke. Outer CMU (Partially Grouted Cell@800 c-c)
  - 16 x 0.25 x (3.5-0.5) m clear height = **12 kN/m**

5.2. Live Load

5.2.1. Roof Live Loads(Lr)

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

TOTAL Lr as UDL ON BEAMS

- UDL on LB's Along Grid Line-A & C between 2 & 3 = 0.50 x 5.10/2 = **1.275 Kn/m**
- UDL on LB's Along Grid Line-B & C between 1 & 2 = 0.50 x 3.0/2 = **0.75 Kn/m**

5.3. Wind Load(WL)

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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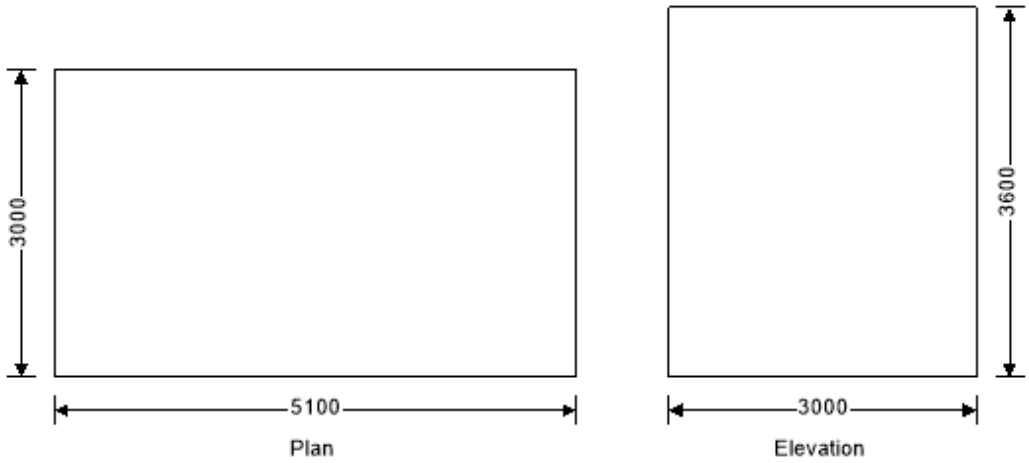
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5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = **5100** mm
Width of building;	d = **3000** mm
Height to eaves;	H = **3600** mm
Mean height;	h = **3600** mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)
Exposure category;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)

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Gust effect factor; G_f = **0.85**(As per SBC-301, Section-7.2.7.1)

Importance Factor; 1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3600	1.03	1.29
3600	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.60	0.80	1.29	0.17	18.36	3.08
B	3.60	-0.50	1.29	-1.26	18.36	-23.07
C	3.60	-0.70	1.29	-1.48	10.80	-15.93
D	3.60	-0.70	1.29	-1.48	10.80	-15.93

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Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.60	0.80	1.29	1.58	18.36	29.10
B	3.60	-0.50	1.29	0.16	18.36	2.96
C	3.60	-0.70	1.29	-0.06	10.80	-0.63
D	3.60	-0.70	1.29	-0.06	10.80	-0.63

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.60	0.80	1.29	0.17	10.80	1.81
B	3.60	-0.36	1.29	-1.10	10.80	-11.91
C	3.60	-0.70	1.29	-1.48	18.36	-27.09
D	3.60	-0.70	1.29	-1.48	18.36	-27.09

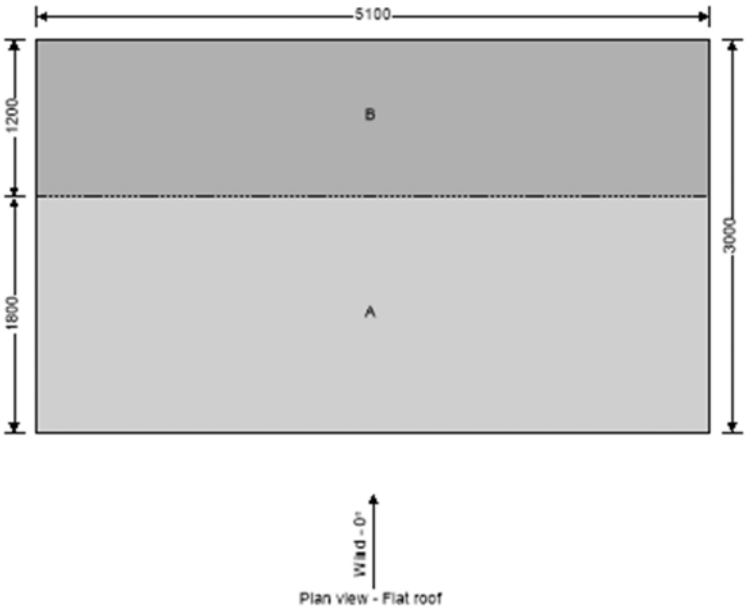
Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

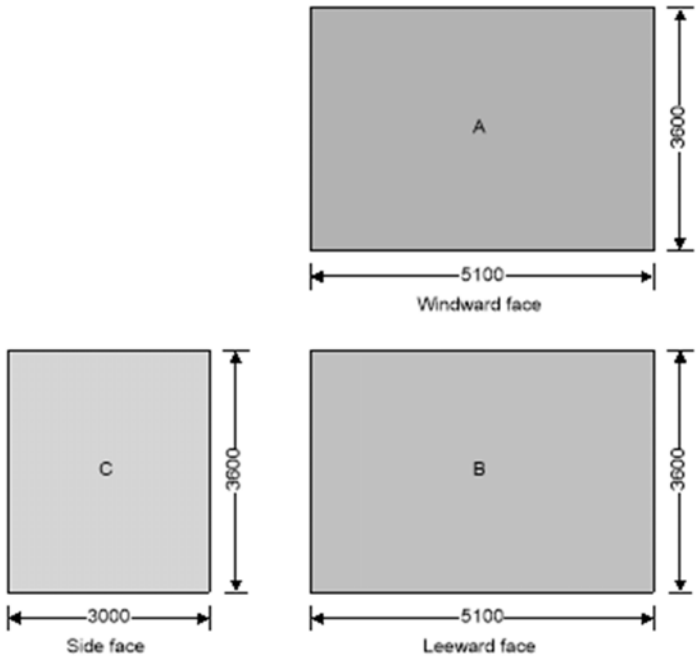
Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.60	0.80	1.29	1.58	10.80	17.12
B	3.60	-0.36	1.29	0.31	10.80	3.40
C	3.60	-0.70	1.29	-0.06	18.36	-1.06
D	3.60	-0.70	1.29	-0.06	18.36	-1.06

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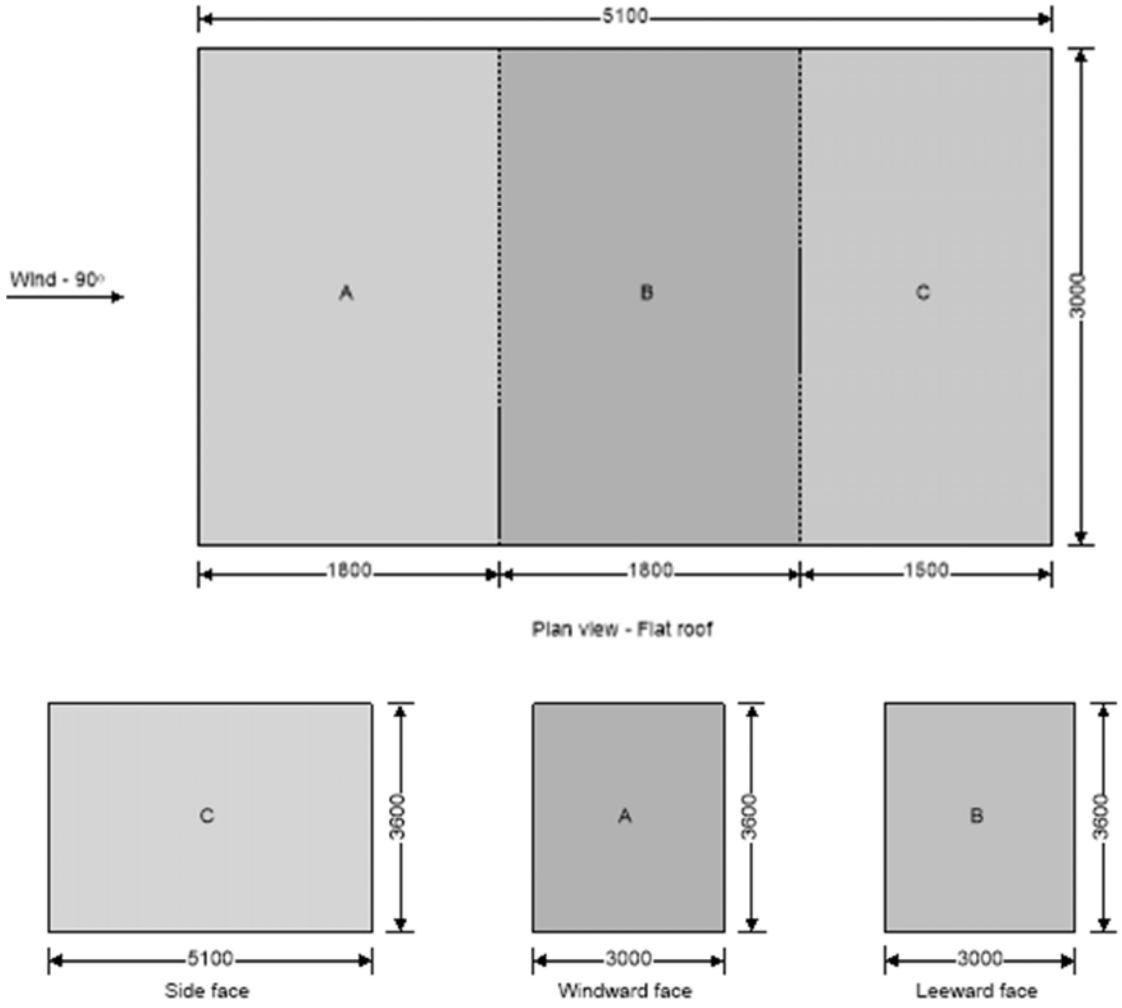




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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.5. Seismic Load(E)

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 Sec

S_{DS} Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

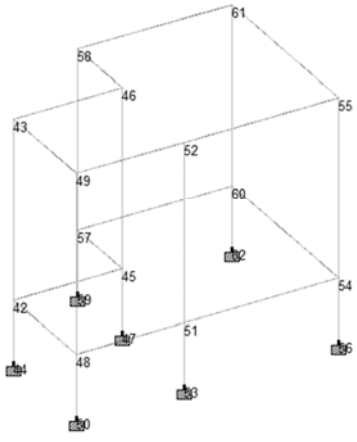
8. DEFLECTIONS

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



WEIGH BRIDGE CONTROL ROOM.std - Node Displacement

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad	Y-Rotan rad	Z-Rotan rad
52	46	0.332	-0.073	-5.056	5.067	-0.001	0.000	-0.000
52	42	0.344	-0.043	-4.960	4.972	-0.001	0.000	-0.000
52	19	0.042	-0.045	3.929	3.929	0.001	-0.000	-0.000
52	40	0.243	-0.073	-3.861	3.869	-0.001	0.000	-0.000
52	23	0.031	-0.075	3.833	3.833	0.001	-0.000	-0.000
55	46	0.338	-0.148	-3.819	3.837	-0.001	-0.000	-0.000
61	46	-0.011	-0.202	-3.809	3.814	0.001	0.000	0.000
55	42	0.349	-0.082	-3.753	3.770	-0.001	-0.000	-0.000
61	42	-0.010	-0.129	-3.745	3.747	0.000	0.000	0.000
61	19	0.295	-0.089	3.707	3.720	0.001	-0.000	-0.000
55	19	0.049	-0.122	3.704	3.706	-0.000	0.000	-0.000
61	23	0.294	-0.162	3.644	3.659	0.001	-0.000	0.000
55	23	0.037	-0.189	3.638	3.643	-0.001	0.000	0.000
49	46	0.329	-0.064	-3.333	3.350	-0.001	0.001	-0.000
43	46	-0.646	-0.153	-3.329	3.395	0.000	0.000	0.000
49	42	0.340	-0.023	-3.281	3.299	-0.000	0.001	-0.000
43	42	-0.641	-0.108	-3.278	3.342	-0.000	0.000	0.000
				0.881	0.000	0.001	-0.035	0.035
				1.763	-0.000	0.001	-0.004	0.004
				2.644	-0.000	0.001	0.030	0.030
				3.525	0.000	0.000	0.000	0.000
			2 EQK LOAD	0.000	0.000	0.000	0.000	0.000
				0.881	0.000	0.000	0.037	0.000
				1.763	0.000	0.010	-0.000	0.010
				2.644	0.000	-0.022	-0.001	0.022
				3.525	0.000	0.000	0.000	0.000

Maximum Attained Displacement as per Analysis is 5mm<<<H/100=3500/100=35mm.

Hence Structure is safe against Drift check.

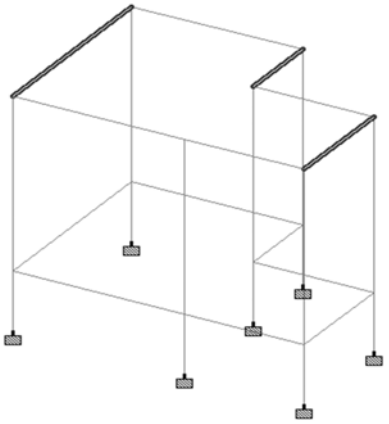
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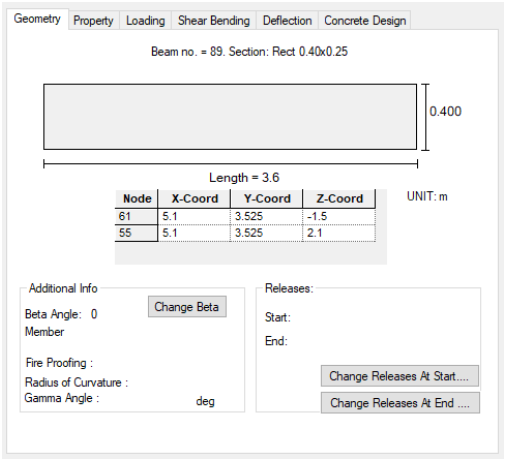
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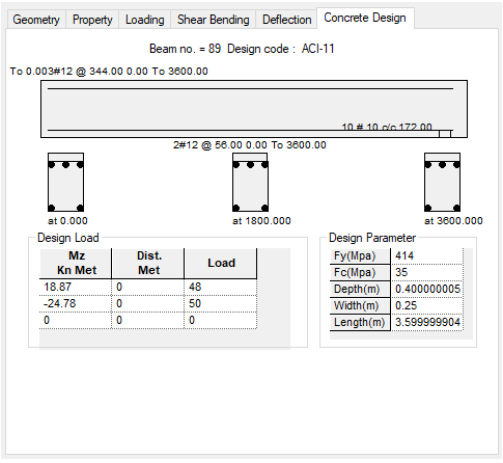
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9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)







Design of Ledger Beam (LB1)

Hence Provide 3-12 dia-Full Straight Bars Top & 3-12 dia-Full Straight Bottom

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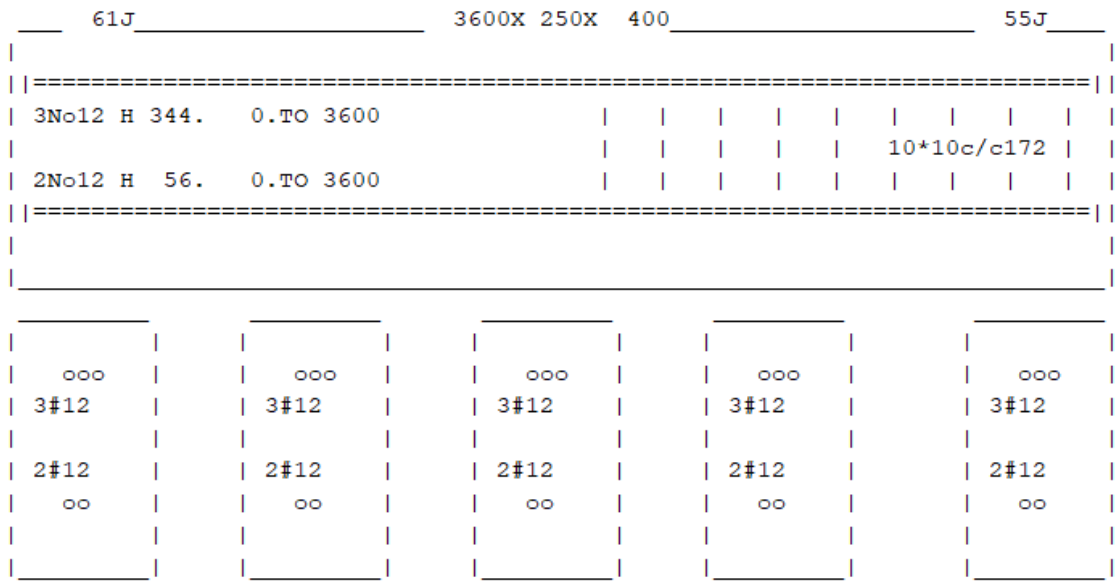
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9.1.1. Staad Output For LB1

```
                B E A M   N O.    89 D E S I G N   R E S U L T S - SHEAR


AT START SUPPORT - Vu=    18.10 KNS   Vc=    87.01 KNS   Vs=     0.00 KNS
Tu=     0.01 KN-MET   Tc=     2.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                    STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    32.07 KNS   Vc=    84.14 KNS   Vs=     0.00 KNS
Tu=     0.00 KN-MET   Tc=     2.9 KN-MET   Ts=     0.0 KN-MET   LOAD     28
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 172. MM  C/C FOR 1460. MM
```

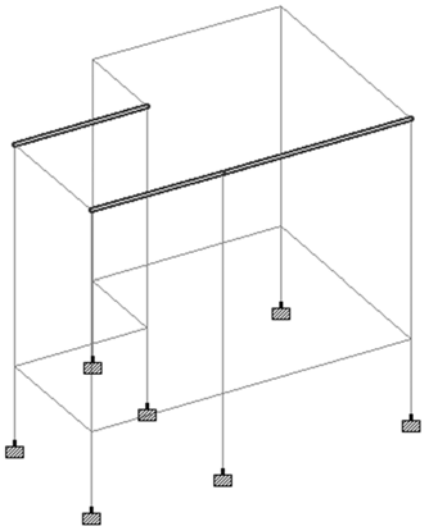


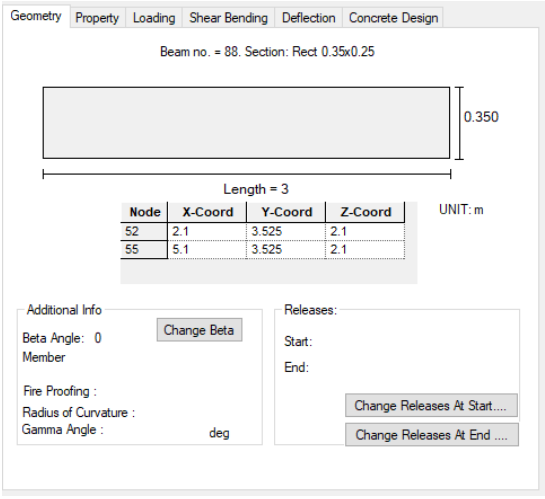
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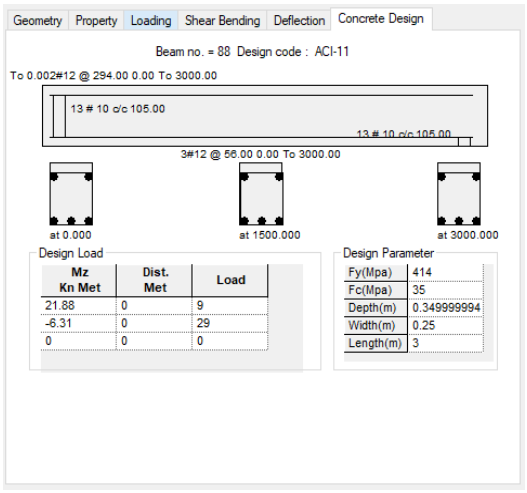
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9.2. Design of Beam-(B1)







Provide 3-12 dia-Full Straight Bars Top & Bottom

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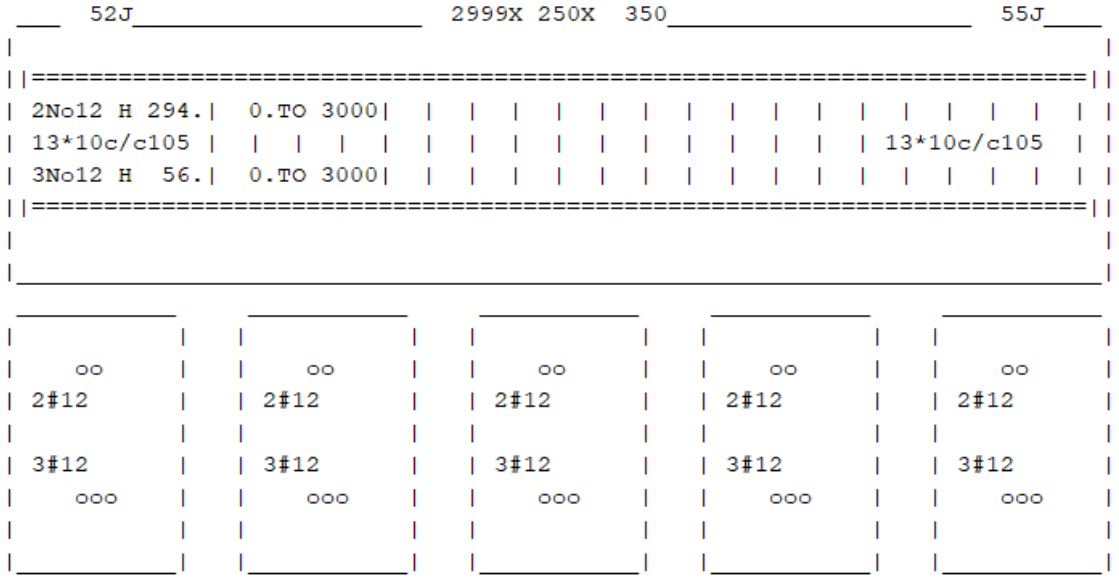
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9.2.1. Staad Output For B1

```
        B E A M  N O.    88 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=     3.40 KNS  Vc=    70.48 KNS  Vs=     0.00 KNS
 Tu=     2.38 KN-MET  Tc=     2.3 KN-MET  Ts=     3.2 KN-MET  LOAD     28
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 105. MM  C/C FOR 1210. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    0.91 SQ.CM.

 AT END   SUPPORT - Vu=     2.59 KNS  Vc=    70.48 KNS  Vs=     0.00 KNS
 Tu=     2.38 KN-MET  Tc=     2.3 KN-MET  Ts=     3.2 KN-MET  LOAD     28
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 105. MM  C/C FOR 1210. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    0.91 SQ.CM.
```

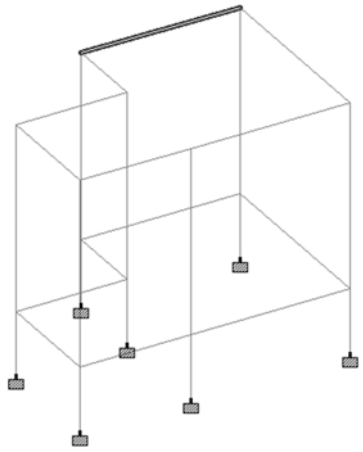


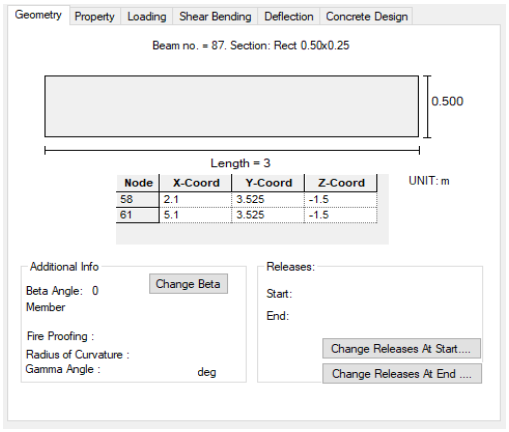
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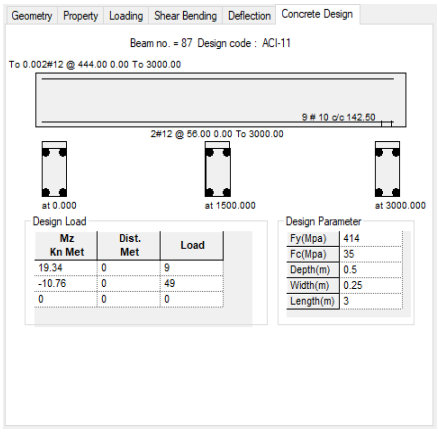
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9.3. Design of Beam-(B2)







Provide minimum 3-16 dia-Full Straight Bars Top & Bottom considering the torsion from the Cantilever Steel Canopy.

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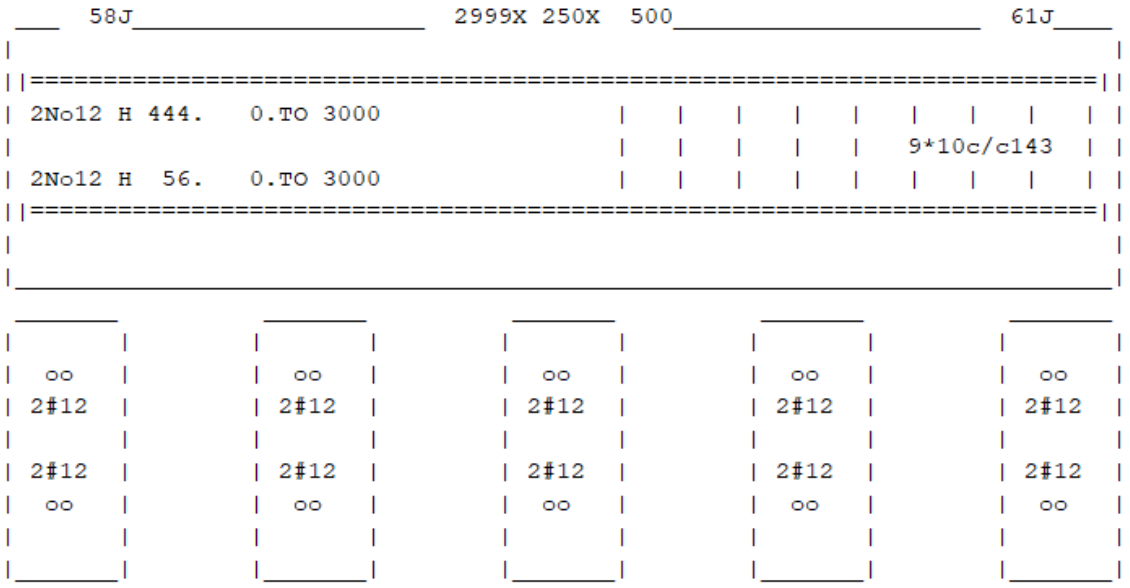
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9.3.1. Staad Output For B2

```
            B E A M   N O.    87 D E S I G N   R E S U L T S - SHEAR


   AT START SUPPORT - Vu=     3.31 KNS   Vc=  191.03 KNS   Vs=     0.00 KNS
   Tu=     0.70 KN-MET   Tc=     3.8 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                       STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=     4.10 KNS   Vc=  114.42 KNS   Vs=     0.00 KNS
   Tu=     5.05 KN-MET   Tc=     3.8 KN-MET   Ts=     6.7 KN-MET   LOAD     25
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1060. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.66 SQ.CM.
```



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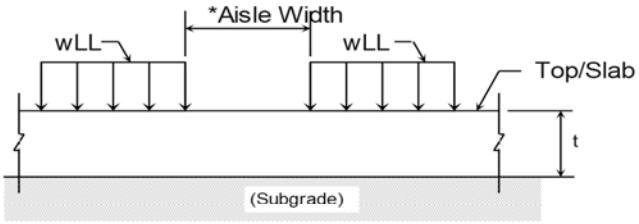
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm	
Concrete Strength, f 'c =	5000	psi	35.000	kpa	
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)
Factor of Safety, FS =	1.500		1.500		
Uniform Live Load, wLL =	104.45	psf	5.00	kpa	



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
ow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-6mm Dia@200 c-c Both Ways-TOP

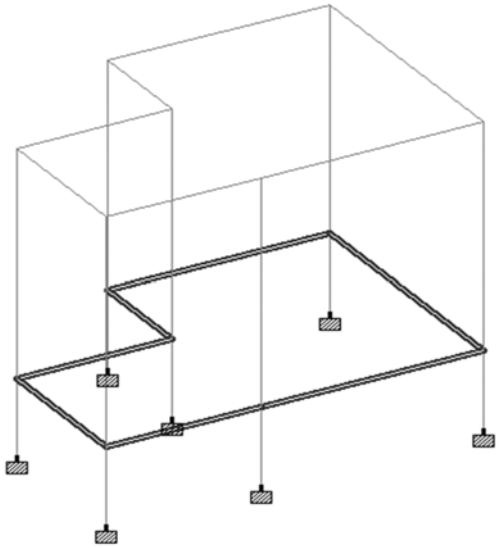
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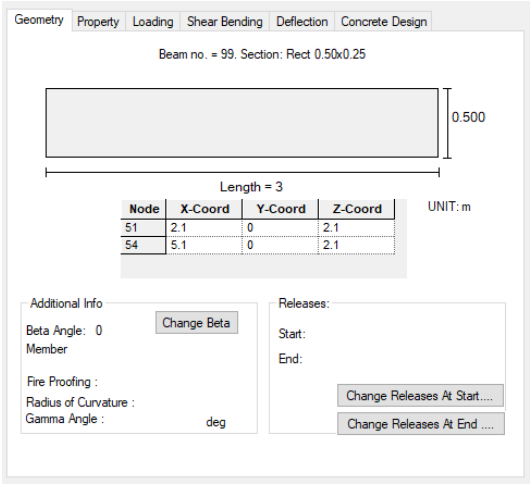
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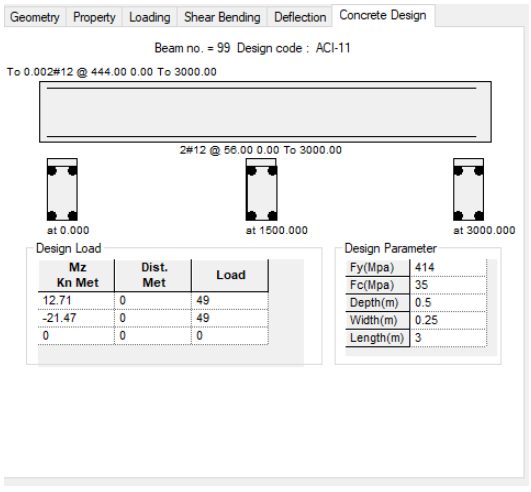
11. DESIGN OF GRADE BEAM

11.1. Grade Beam(GB1)



MOMENT ENVELOPE FOR BEAMS GB1





Provide minimum 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

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11.1.1. STAAD OUTPUT FOR GB1

```
            B E A M  N O.    99 D E S I G N  R E S U L T S - SHEAR

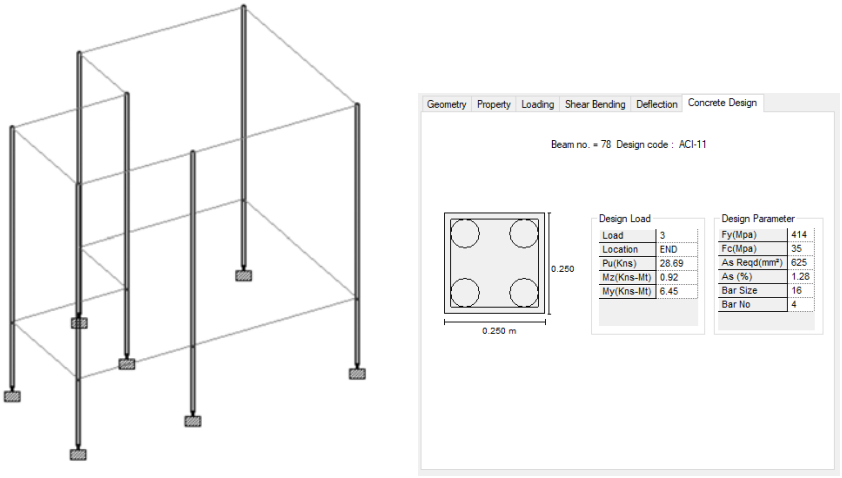

   AT START SUPPORT - Vu=   18.25 KNS  Vc=  109.77 KNS  Vs=    0.00 KNS
   Tu=    0.69 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                    STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=   13.43 KNS  Vc=  111.42 KNS  Vs=    0.00 KNS
   Tu=    0.69 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                    STIRRUPS ARE NOT REQUIRED.
 ___  51J_____________________  2999X 250X  500_____________________  54J____
|                                                                             |
||=============================================================================||
| 2No12 H 444.   0.TO 3000                                                    |
|                                                                             |
| 2No12 H  56.   0.TO 3000                                                    |
||=============================================================================||
|                                                                             |
|_____________________________________________________________________________|
 _______        _______        _______        _______        _______
|       |      |       |      |       |      |       |      |       |
|  oo   |      |  oo   |      |  oo   |      |  oo   |      |  oo   |
| 2#12  |      | 2#12  |      | 2#12  |      | 2#12  |      | 2#12  |
|       |      |       |      |       |      |       |      |       |
| 2#12  |      | 2#12  |      | 2#12  |      | 2#12  |      | 2#12  |
|  oo   |      |  oo   |      |  oo   |      |  oo   |      |  oo   |
|       |      |       |      |       |      |       |      |       |
|_______|      |_______|      |_______|      |_______|      |_______|
```


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12. CONCRETE COLUMN DESIGN



Column's C1

12.1. Staad Output For Column C1

```
      ACI 318-11    COLUMN  NO.     78    DESIGN RESULTS
      ==================================================


FY - 413.7   FC -   35.0 MPA,   SQRE SIZE - 250.0 X 250.0 MMS, TIED
         ONLY MINIMUM STEEL IS REQUIRED.
         AREA OF STEEL REQUIRED =    625.0   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-------------------------------------------------------------

  4 - 16 MM                 1.287         3       END      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    10 SPACING 250.00 MM
```


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13. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 570 Kpa. (Refer Doc-MD-216-3000-EG-CV-CAL-1008-A01 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Profile / Borehole	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement
				Settlement	Shear Failure		
-	-	(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)
BH-01	Square Isolated	1.20	1.0 x 1.0	1500	570	570	25
		1.50		2000	850	850	25
		1.20	2.0 x 2.0	1000	750	750	25
		1.50		1200	980	980	25
		1.20	3.0 x 3.0	775	900	775	25
		1.50		950	1120	950	25
		1.20	4.0 x 4.0	625	1040	625	25
		1.50		750	1250	750	25
		1.20	5.0 x 5.0	525	1172	525	25
		1.50		600	1342	600	25

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.) BUILDING PLANS	MD-216-3000-EG-ME-GAD-1002
2.) LAYOUT & DETAIL OF BUILDING STRUCTURE	MD-216-4000-EG-CV-GAD-1084
3.) RC DETAILS BUILDING STRUCTURE	MD-216-4000-EG-CV-DED-1085
4.) GEOTECHNICAL REPORT	MD-216-3000-EG-CV-CAL-1008
5.) STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

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```
SSTAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
42 0 0 0; 43 0 3.525 0; 44 0 -1.2 0; 45 2.1 0 0; 46 2.1 3.525 0; 47 2.1 -1.2 0;
48 0 0 2.1; 49 0 3.525 2.1; 50 0 -1.2 2.1; 51 2.1 0 2.1; 52 2.1 3.525 2.1;
53 2.1 -1.2 2.1; 54 5.1 0 2.1; 55 5.1 3.525 2.1; 56 5.1 -1.2 2.1;
57 2.1 0 -1.5; 58 2.1 3.525 -1.5; 59 2.1 -1.2 -1.5; 60 5.1 0 -1.5;
61 5.1 3.525 -1.5; 62 5.1 -1.2 -1.5;
MEMBER INCIDENCES
70 42 43; 71 42 44; 72 45 46; 73 45 47; 74 48 49; 75 48 50; 76 51 52; 77 51 53;
78 54 55; 79 54 56; 80 57 58; 81 57 59; 82 60 61; 83 60 62; 84 43 46; 85 49 52;
86 43 49; 87 58 61; 88 52 55; 89 61 55; 90 60 54; 91 58 46; 93 57 45; 95 42 48;
96 42 45; 97 48 51; 98 57 60; 99 51 54;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 28000
G 9.28139e+006
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
70 TO 83 PRIS YD 0.25 ZD 0.25
MEMBER PROPERTY AMERICAN
90 93 95 TO 99 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
86 89 91 PRIS YD 0.4 ZD 0.25
MEMBER PROPERTY AMERICAN
84 85 88 PRIS YD 0.35 ZD 0.25
MEMBER PROPERTY AMERICAN
87 PRIS YD 0.5 ZD 0.25
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
44 47 50 53 56 59 62 FIXED
******************************************************************************
MEMBER CRACKED
84 TO 91 93 95 TO 99 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
70 TO 83 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 35.000 M 25.000 M 2.000 0.010 0 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.59699 1.59699 HEIG  0  4.572
```

```
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 6.000 M 6.000 M 2.000 0.010 1 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.26591 -1.26591 HEIG  0  4
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 6.000 M 6.000 M 2.000 0.010 2 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.898 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.53032 -1.53032 HEIG  0  4
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 6.000 M 6.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.898 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.64692 1.64692 HEIG  0  4.572
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 6.000 M 6.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.898 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.29711 -1.29711 HEIG  0  4
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 6.000 M 6.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.898 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.53032 -1.53032 HEIG  0  4
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
87 UMOM GX -2
90 93 95 TO 99 UNI GY -12
86 UNI GY -14.84
91 UNI GY -8
89 UNI GY -14.84 2.1 3.6
89 UNI GY -8 0 2.1
LOAD 4 LOADTYPE Live  TITLE LL
MEMBER LOAD
```

86 89 91 UNI GY -1.25
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 4
WIND LOAD -X 1 TYPE 2 YR 0 4
WIND LOAD -Z 1 TYPE 3 YR 0 4
WIND LOAD -Z -1 TYPE 3 YR 0 4
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 XR 0 75 YR 0 4
WIND LOAD -Z 1 TYPE 5 XR 0 75 YR 0 4
WIND LOAD -X 1 TYPE 6 XR 0 75 YR 0 4
WIND LOAD -X -1 TYPE 6 XR 0 75 YR 0 4
LOAD 7 LOADTYPE Wind TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 4
WIND LOAD -X -1 TYPE 2 YR 0 4
WIND LOAD -Z 1 TYPE 3 YR 0 4
WIND LOAD -Z -1 TYPE 3 YR 0 4
LOAD 8 LOADTYPE Wind TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 4
WIND LOAD -Z -1 TYPE 5 YR 0 4
WIND LOAD -X 1 TYPE 6 YR 0 4
WIND LOAD -X -1 TYPE 6 YR 0 4
LOAD 9 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
70 72 74 76 78 80 82 84 TO 89 91 TEMP 25 25 25
LOAD COMB 10 SLS: (DL)
3 1.0
LOAD COMB 11 SLS: (DL+T)
3 1.0
LOAD COMB 12 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 13 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 14 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 15 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 16 SLS: (D+0.75WX1+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 17 SLS: (D+0.75WZ1+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 18 SLS: (0.6D+WX1)
3 0.6 5 1.0
LOAD COMB 19 SLS: (0.6D+WZ1)
3 0.6 6 1.0
LOAD COMB 20 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 21 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 22 SLS: (D+WX1)
3 1.0 5 1.0
LOAD COMB 23 SLS: (D+WZ1)
3 1.0 6 1.0
LOAD COMB 24 ULS: (1.4 DL)
3 1.4
LOAD COMB 25 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 26 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 27 ULS: (1.2D+0.80WX1+1.6LR)

3 1.2 5 0.8 4 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 29 ULS: (1.2D+1.6WX1+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 30 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 31 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 32 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 33 ULS: (0.90D+1.6WX1)
3 0.9 5 1.6
LOAD COMB 34 ULS: (0.90D+1.6WZ1)
3 0.9 6 1.6
LOAD COMB 35 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 36 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 39 SLS: (D+0.75WX2+0.75LR)
3 1.0 7 0.75 4 0.75
LOAD COMB 40 SLS: (D+0.75WZ2+0.75LR)
3 1.0 8 0.75 4 0.75
LOAD COMB 41 SLS: (0.6D+WX2)
3 0.6 7 1.0
LOAD COMB 42 SLS: (0.6D+WZ2)
3 0.6 8 1.0
LOAD COMB 45 SLS: (D+WX2)
3 1.0 7 1.0
LOAD COMB 46 SLS: (D+WZ2)
3 1.0 8 1.0
LOAD COMB 47 ULS: (1.2D+0.80WX2+1.6LR)
3 1.2 7 0.8 4 1.6
LOAD COMB 48 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 8 1.6 4 0.5
LOAD COMB 49 ULS: (1.2D+1.6WX2+0.5LR)
3 1.2 4 0.5 7 1.6
LOAD COMB 50 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 4 0.5 8 1.6
LOAD COMB 53 ULS: (0.90D+1.6WX2)
3 0.9 7 1.6
LOAD COMB 54 ULS: (0.90D+1.6WZ2)
3 0.9 8 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 ALL
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
MINSEC 10 ALL
MINMAIN 16 MEMB 70 TO 83
DESIGN COLUMN 70 TO 83
DESIGN BEAM 84 TO 91 93 95 TO 99
END CONCRETE DESIGN
FINISH
FINISH

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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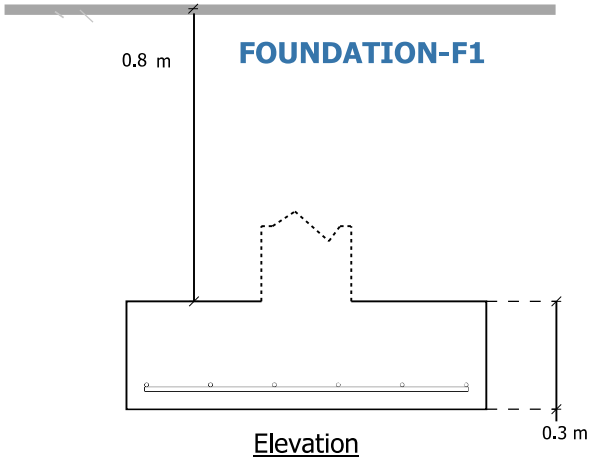
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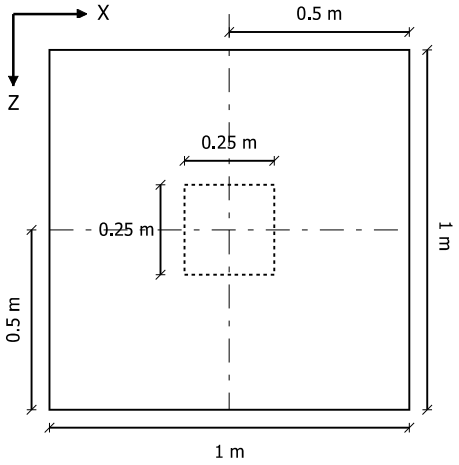
Isolated Footing Design(ACI 318-08)

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
59	1	1.000m	1.000m	0.300m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
59	6 - ϕ12	6 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 300.000mm
Footing Length - X (Fl) : 1000.000mm
Footing Width - Z (Fw) : 1000.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.250m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 200.000mm
Maximum Bar Spacing : 200.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.000m

Initial Width (W_o) = 1.000m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	58.463	-5.506	-1.070	-0.427	2.082
11	58.463	-5.506	-1.070	-0.427	2.082
12	59.309	-5.508	-1.059	-0.425	2.082
13	59.098	-5.508	-1.062	-0.425	2.082
14	55.746	-5.464	-0.474	0.019	2.046
15	57.211	-4.873	-1.078	-0.423	1.582
16	59.596	-1.374	-4.492	-2.683	-0.981
17	40.137	-4.403	4.612	3.606	1.315
20	33.825	-2.670	-0.650	-0.252	0.749
21	32.361	-3.262	-0.046	0.190	1.213
22	59.128	0.006	-5.643	-3.437	-2.001
23	33.182	-4.033	6.495	4.948	1.060
39	71.887	-10.254	-4.446	-2.742	5.591
40	76.159	-4.678	-6.114	-4.056	1.585

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
45	75.515	-11.834	-5.582	-3.515	6.761
46	81.212	-4.399	-7.807	-5.267	1.420

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	81.849	-7.708	-1.498	-0.597	2.915
25	82.271	-7.709	-1.493	-0.596	2.915
26	82.271	-7.709	-1.493	-0.596	2.915
27	72.041	-2.202	-4.925	-2.917	-0.769
28	30.129	-4.251	10.825	8.088	0.862
29	71.642	2.210	-8.596	-5.327	-4.035
30	30.129	-4.251	10.825	8.088	0.862
31	68.366	-5.702	-1.295	-0.507	1.784
32	66.274	-6.547	-0.433	0.124	2.447
33	53.680	3.863	-8.280	-5.200	-4.660
34	12.167	-2.598	11.141	8.215	0.238
35	48.735	-4.895	-0.112	0.252	1.823
36	50.827	-4.051	-0.974	-0.379	1.160
47	85.151	-11.673	-4.876	-2.980	6.241
48	106.977	-4.838	-12.057	-8.256	1.439
49	97.862	-16.733	-8.497	-5.452	9.984
50	106.977	-4.838	-12.057	-8.256	1.439
53	79.900	-15.080	-8.182	-5.325	9.360
54	89.015	-3.185	-11.742	-8.129	0.814

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000m^2

Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 0.512m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.000 m Governing Load Case : # 10

Width (W_2) = 1.000 m Governing Load Case : # 10

Depth (D_2) = 0.300 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.000 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 7.200 kN

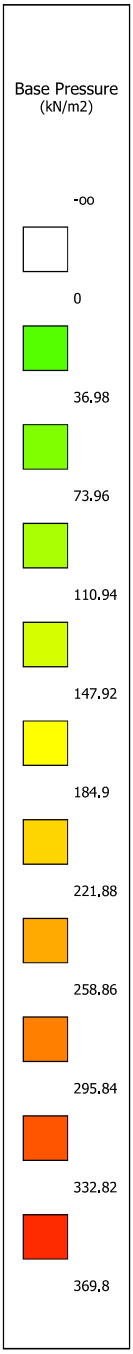
Gross Soil Bearing Capacity = 369.80 kN/m2

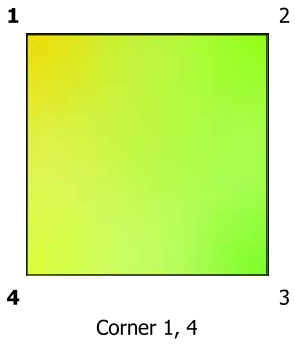
Soil Weight On Top Of Footing = 13.500 kN

Pressures at 4 Corners

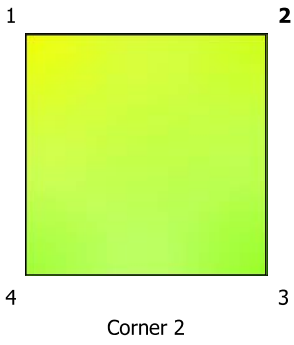
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

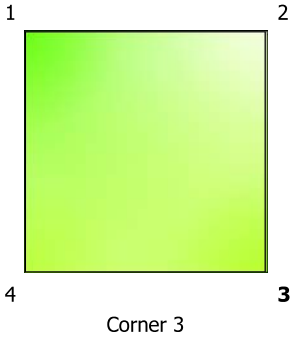




Corner 1, 4

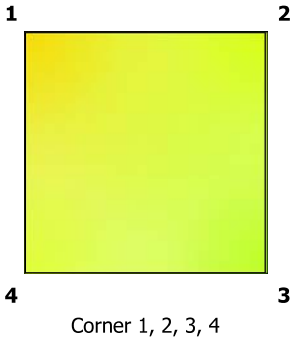


Corner 2



Corner 3

Governing Base Pressure Contour



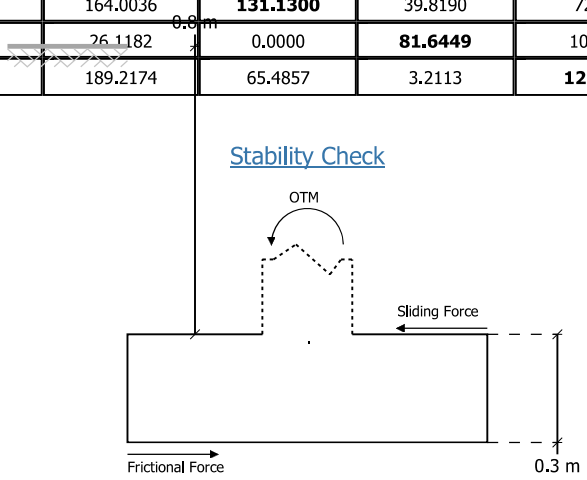
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**189.2174**	65.4857	3.2113	126.9430	0.000
46	164.0036	**131.1300**	39.8190	72.6926	0.000
23	26.1184	-1.1135	**81.6444**	108.8763	0.000
45	189.2174	65.4857	3.2113	**126.9430**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**189.2174**	65.4857	3.2113	126.9430
46	164.0036	**131.1300**	39.8190	72.6926
23	26.1182	0.0000	**81.6449**	108.8768
45	189.2174	65.4857	3.2113	**126.9430**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	7.189	36.991	9.980	1.500	53.245	10.661	1.500
11	7.189	36.991	9.980	1.500	53.245	10.661	1.500
12	7.262	37.771	10.086	1.500	54.177	10.773	1.500
13	7.244	37.574	10.060	1.500	53.943	10.745	1.500
14	6.995	80.602	9.856	1.500	311.622	10.432	1.500
15	7.995	36.136	11.039	1.500	52.502	12.872	1.500
16	29.216	8.938	12.087	1.500	10.017	71.022	1.500
17	6.909	6.596	6.747	1.500	6.142	11.625	1.500
20	10.209	41.943	14.028	1.500	61.463	17.730	1.500
21	8.134	574.219	11.502	1.500	152.312	12.206	1.500
22	7040.662	7.073	10.002	1.500	7.825	20.037	1.500
23	6.681	4.148	4.984	1.500	3.939	11.971	1.500
39	4.515	10.413	5.858	1.500	11.415	5.367	1.500
40	10.353	7.921	8.896	1.500	8.260	16.280	1.500
45	4.065	8.619	5.200	1.500	9.313	4.687	1.500
46	11.583	6.527	8.042	1.500	6.726	18.683	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 45
Governing Disturbing Force : -11.834kN
Governing Restoring Force : 48.107kN
Minimum Sliding Ratio for the Critical Load Case : 4.065
Critical Load Case for Overturning about X-Direction : 23
Governing Overturning Moment : 6.896kNm
Governing Resisting Moment : 27.165kNm
Minimum Overturning Ratio for the Critical Load Case : 3.939

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction :	23
Governing Disturbing Force :	6.495kN
Governing Restoring Force :	26.941kN
Minimum Sliding Ratio for the Critical Load Case :	4.148
Critical Load Case for Overturning about Z-Direction :	45
Governing Overturning Moment :	10.311kNm
Governing Resisting Moment :	48.331kNm
Minimum Overturning Ratio for the Critical Load Case :	4.687

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

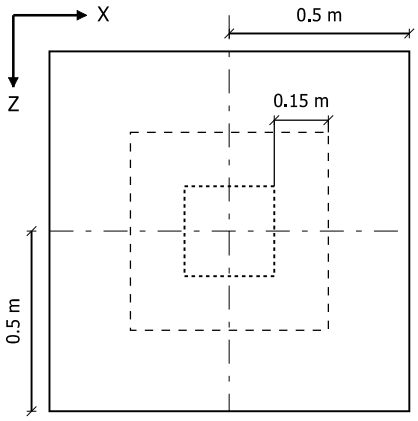
Critical Load Case for Sliding along Resultant Direction :	23
Governing Disturbing Force :	7.645kN
Governing Restoring Force :	38.100kN
Minimum Sliding Ratio for the Critical Load Case :	4.984

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.300m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.244m

For rectangular column, β_c = B_{col} / D_{col} = 1.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 96.519kN, Load Case # 48

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 1.976m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1421.090kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 1643.555kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 947.393kN

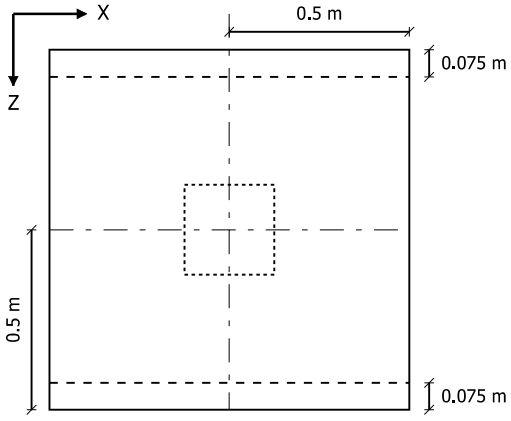
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 710.545kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 239.725kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.131m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

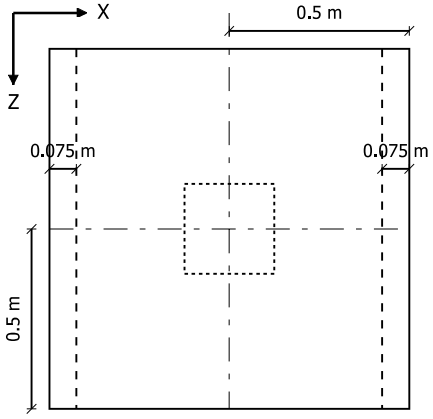
From above calculations, 0.75 X V_c = 179.794 kN

Critical load case for V_{ux} is # 48 $V_{ux} = V_{ux}|_{x=D_x}$ = 24.835 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 239.725 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.131 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

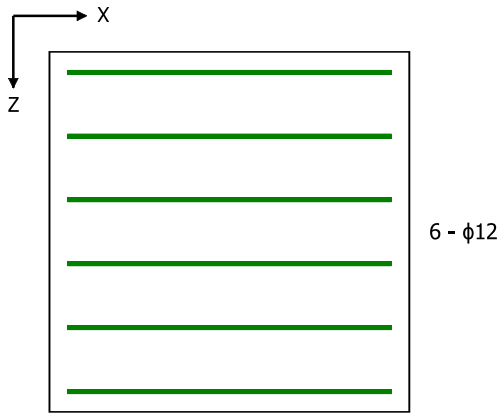
From above calculations, 0.75 X V_c = 179.794 kN

Critical load case for V_{uz} is # 49 $V_{uz} = V_{uz}|_{z=D_z}$ = 25.780 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.244 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00173
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.375 m
Ultimate moment,	$M_u\big\|_{z=D_z}$ =	13.082 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	14.536 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00058
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00047
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	518.712 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 200.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.325	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	6	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	678.579	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.244	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00278	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

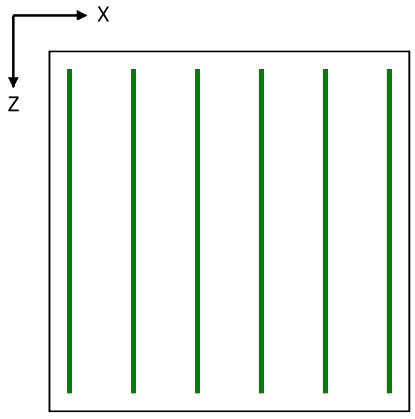
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 200.000mm

Provided Steel Area / Required Steel Area = 1.308

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



6 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 48

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.232	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl.7.12.2, ρ_{min} =		0.00174	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	0.375	m
Ultimate moment,	$M_u\|_{x=D_x}$ =	12.733	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	14.148	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00063	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00049	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	520.708	mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 200.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.325	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	6	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

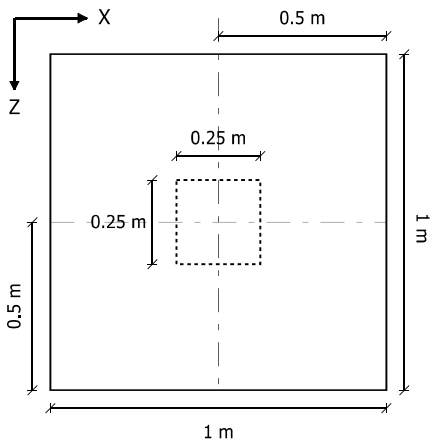
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	678.579	mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.232	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00292	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

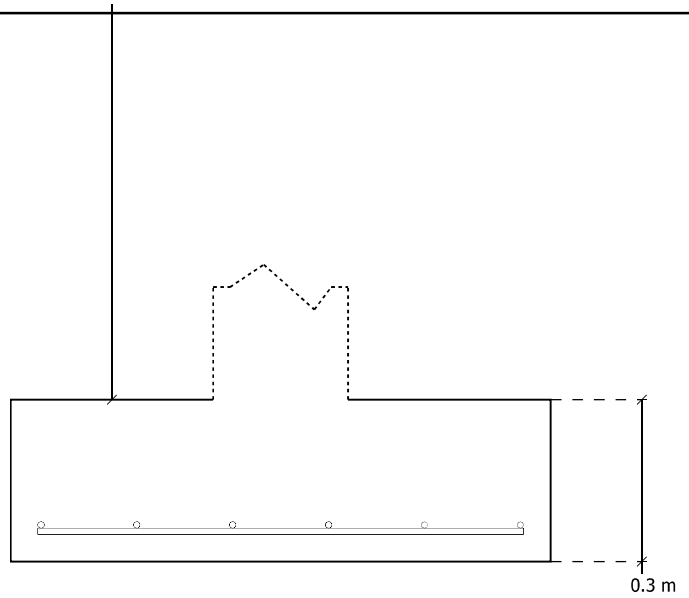
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 200.000mm

Provided Steel Area / Required Steel Area = 1.303

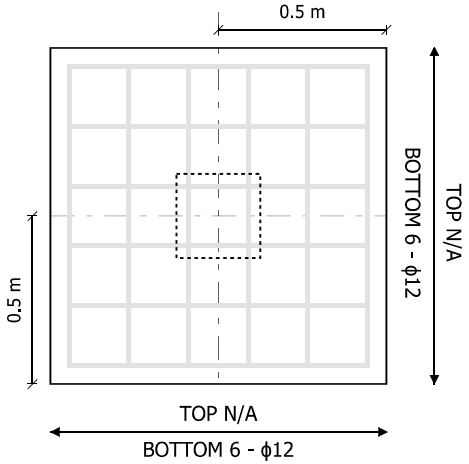
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1 m	1 m	0.3 m	N/A	N/A	6 - ϕ12	6 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet